

Comparing GIS-Based and WUDAPT Approaches for Local Climate Zone Mapping: A Case Study of Denton County, TX

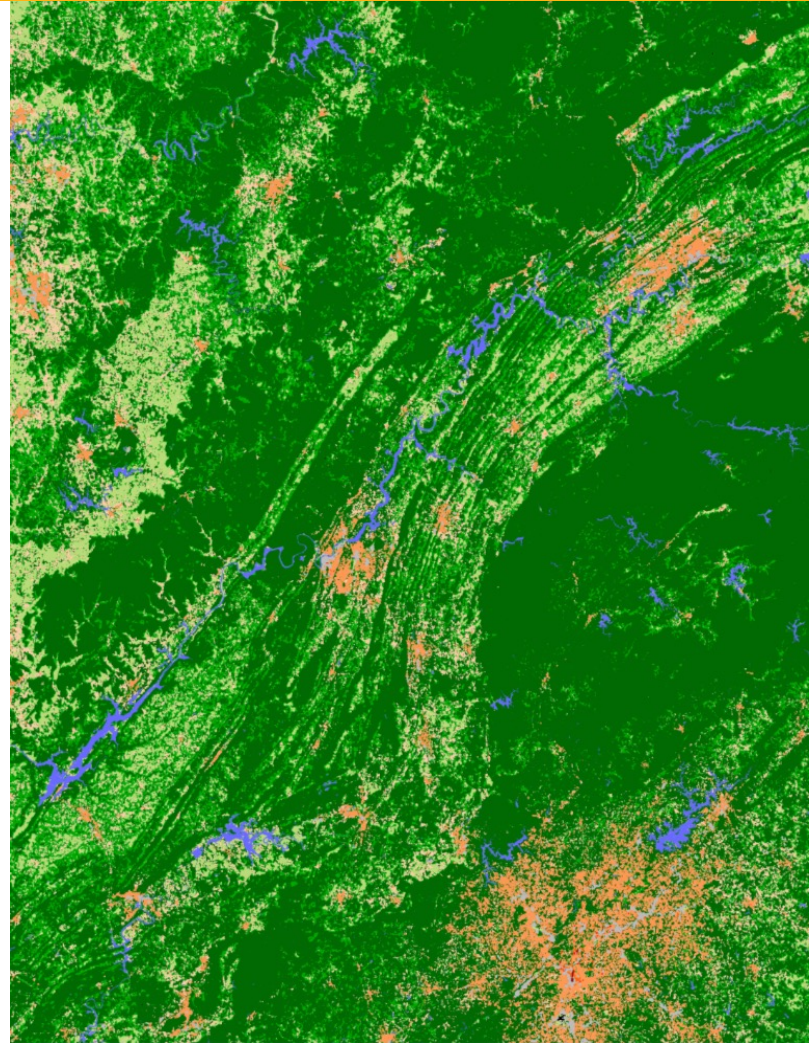
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UC Berkeley

Agenda

- 1 Current Challenges & Local Climate Zone
- 2 GIS Mapping
- 3 Evaluation & Comparison
- 4 Limitations



Challenges in Urban Heat Island Quantification



“urban” has no single, objective meaning,
and climatological relevance

UHI Intensity = temperature in **urban areas** –
temperature in **rural areas**

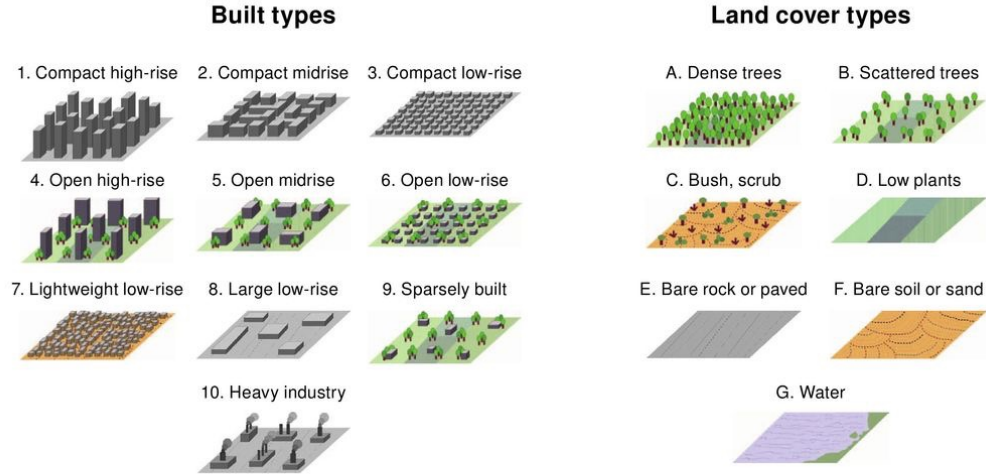
Urban-rural division is not sufficient to
divide cities and countrysides



What is Local Climate Zone (LCZ)?

1. Sky view factor
2. Aspect ratio
3. Building surface fraction
4. Impervious surface fraction
5. Pervious surface fraction
6. Height of roughness
7. Terrain roughness
8. Surface Admittance
9. Surface albedo
10. Anthropogenic heat output

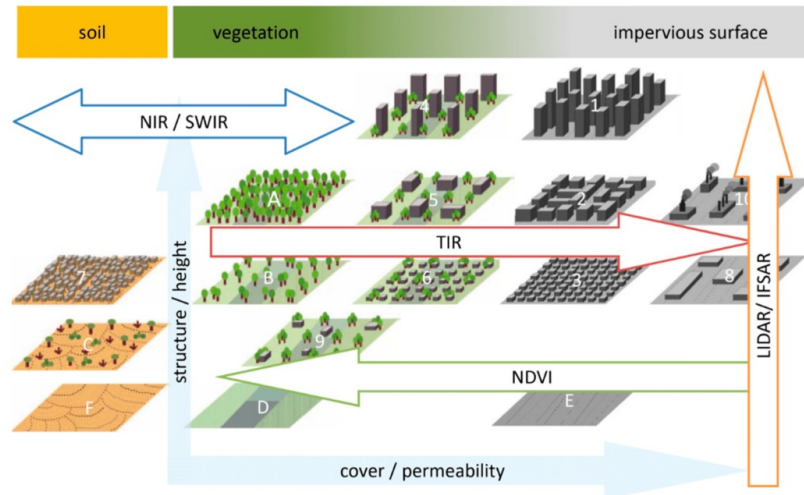
Local Climate Zones



Stewart, I. D., & Oke, T. R. (2012). Local climate zones for urban temperature studies. *Bulletin of the American Meteorological Society*, 93(12), 1879-1900.

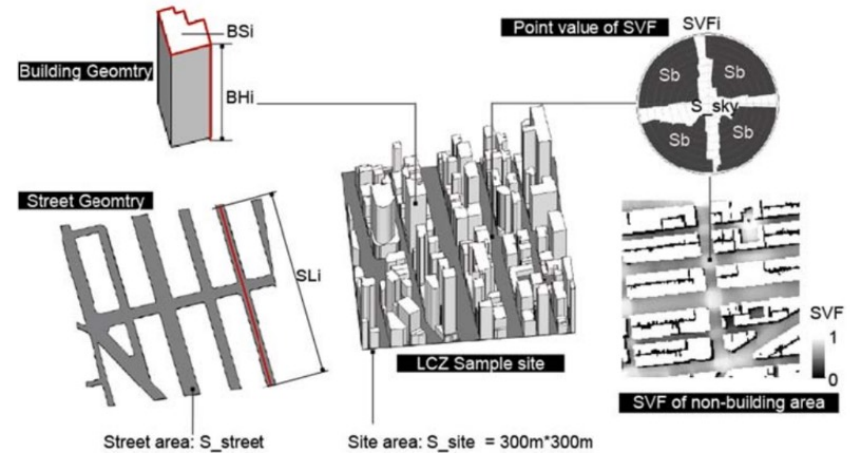
Mainstream LCZ mapping approaches

Machine Learning Based



*Rely on high training sample quality

GIS Based



*High requirement for spatial data

WUDAPT Global LCZ Dataset

A global map of Local Climate Zones

This interface allows you to interactively explore the 100 m spatial resolution global map of Local Climate Zones (LCZ), that is derived from multiple earth observation datasets and expert LCZ class labels. The LCZ typology (Stewart and Oke, 2012) is a universal urban typology that can distinguish urban areas on a holistic basis, accounting for the typical combination of micro-scale land-covers and associated physical properties. The LCZ scheme is distinguished from other land use / land cover schemes by its focus on urban and rural landscape types, which can be described by any of the 17 classes in the LCZ scheme. Its strong added value is thus the diversity of urban classes, which are easily interpretable and globally consistent, capturing the intra-urban variability of surface forms and land functions.

Publication:

[Stewart, I. D., et al. \(2022\)](#)

Demuzere, M., Kitter, J., Martini, A., Mills, G., Mouda, C., Stewart, I. D., van Vliet, J., and Beechel, B. (2022): A global map of local climate zones to support earth system modelling and urban-scale environmental science, *Earth Syst. Sci. Data*, 14, 3839–3873, <https://doi.org/10.5194/essd-14-3839-2022>.

Dataset & Download:

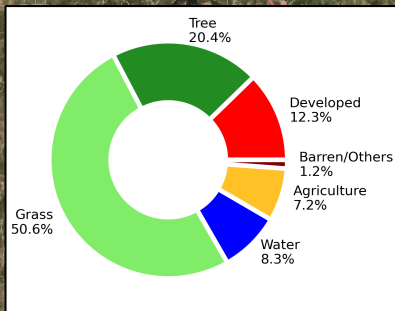
[OpenStreetMap](#)

Google Earth Engine:

[Google Earth Engine](#)



How is its performance compared to a GIS-based method?



Legend

- County Line
- Developed, Medium Intensity
- Developed, High Intensity
- Developed, Low Intensity
- Developed, Open Space



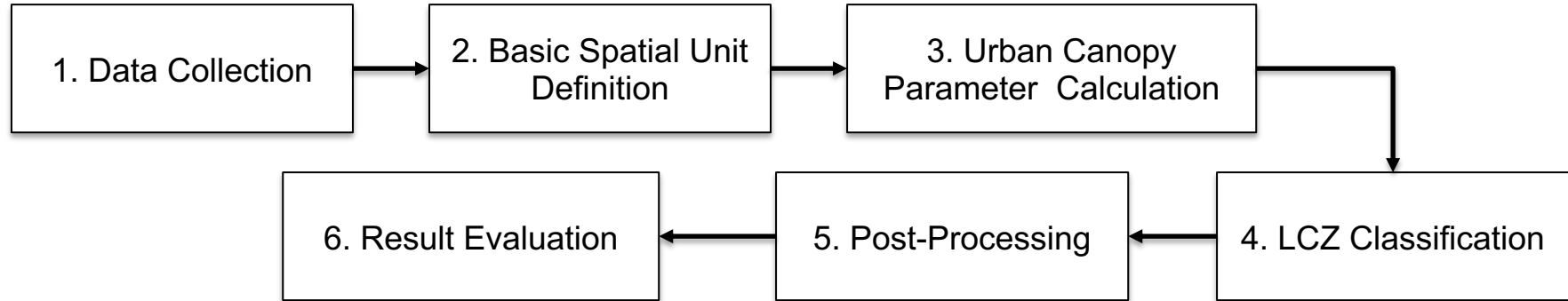
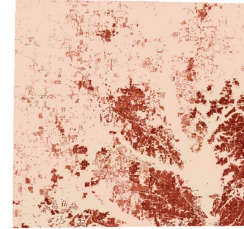
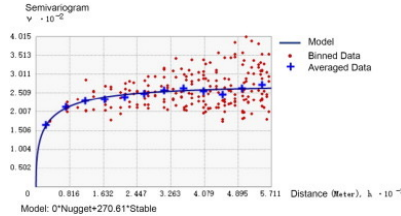
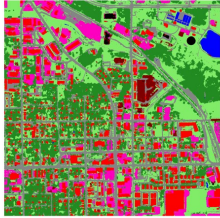
Denton County

Tarrant County

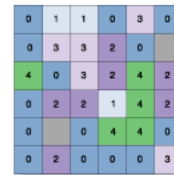
Dallas County



General LCZ mapping Process



OBJECTID	ClassValue	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_10	Total	U_Accuracy	Kappa
1	C_1	7	1	0	0	1	0	0	0	1	0	10	0.7	0
2	C_2	4	46	1	2	0	2	0	0	0	55	0.93554	0	0
3	C_3	0	0	9	0	0	0	0	0	1	0	10	0.9	0
4	C_4	1	0	0	9	0	0	0	0	0	0	10	0.9	0
5	C_5	0	0	0	0	10	0	0	0	0	0	10	1	0
6	C_6	1	0	0	0	0	9	0	0	0	0	10	0.9	0
7	C_7	0	1	2	1	0	0	3	0	3	0	10	0.3	0
8	C_8	0	1	0	0	0	0	0	9	0	0	10	0.9	0
9	C_9	0	0	0	0	0	0	0	0	15	0	15	1	0
10	C_10	0	1	3	0	0	0	0	0	0	6	10	0.6	0
11	Total	13	50	15	12	11	11	3	9	20	6	150	0	0
12	P_Accuracy	0.538462	0.92	0.6	0.75	0.909091	0.818182	1	1	0.75	1	0	0.82	0
13	Kappa	0	0	0	0	0	0	0	0	0	0	0	0.762042	0

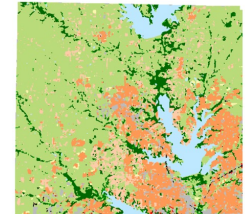


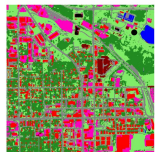
=



InRas1

OutRas

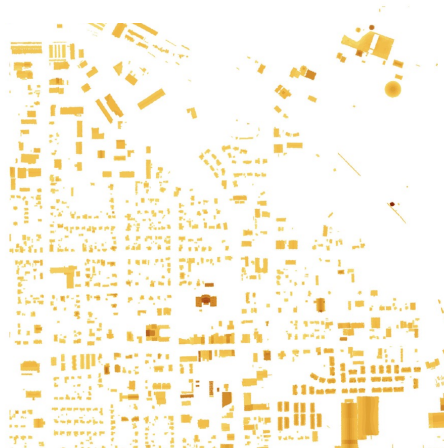




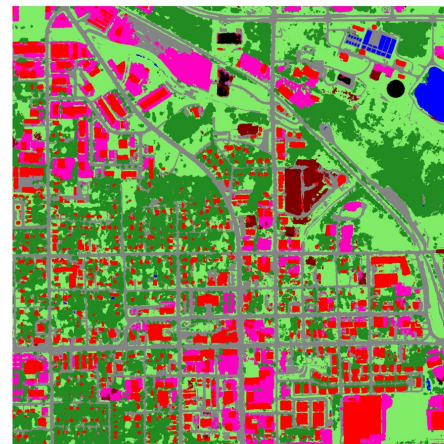
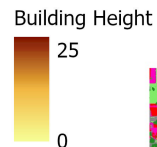
Data Collection



Microsoft Building Polygons



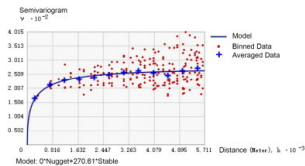
Derive from
LiDAR tiles



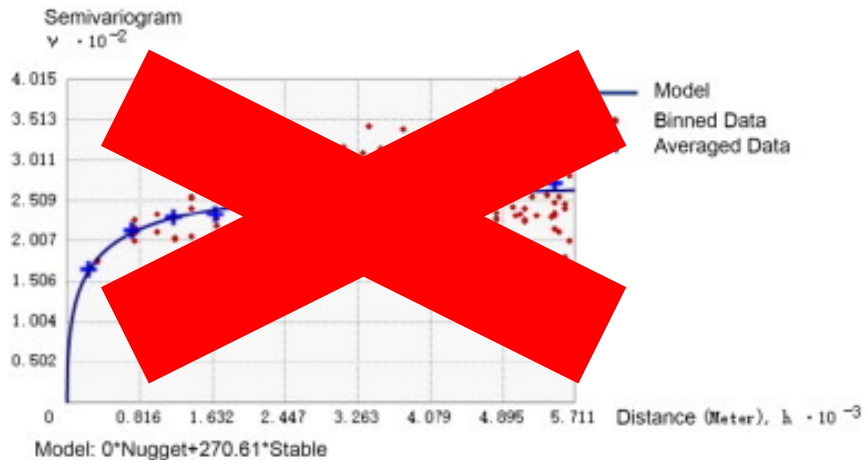
UrbanWatch database

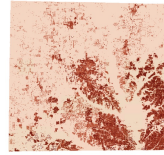
LULC

-  Building
-  Tree canopy
-  Grass
-  Road
-  Water
-  Agriculture
-  Parking lot
-  Barren
-  Others

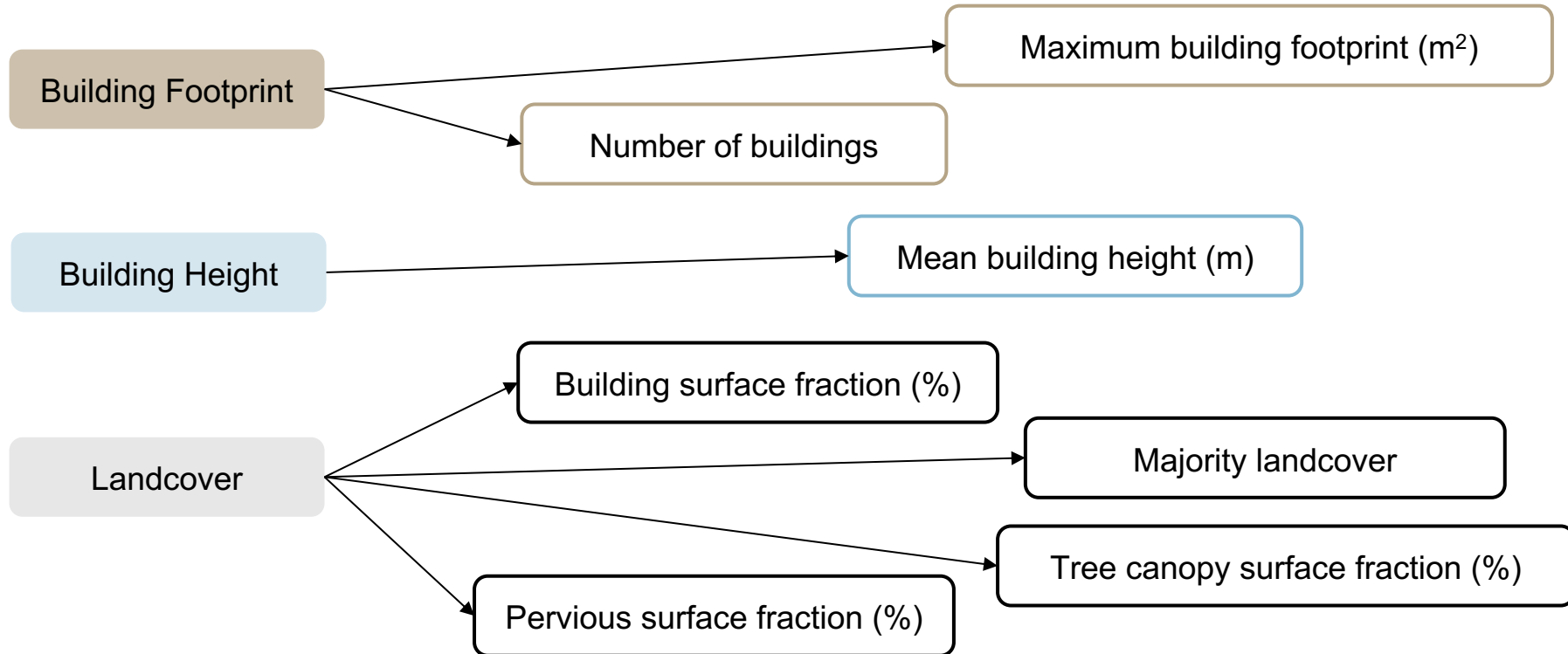


Spatial autocorrelation of building height





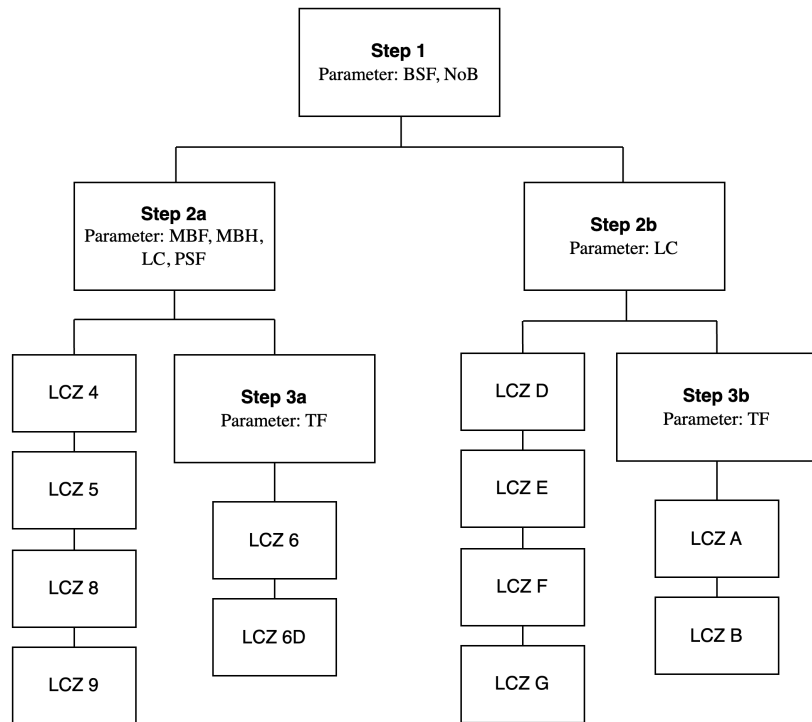
Urban Canopy Parameter

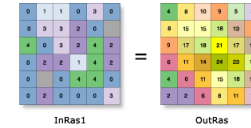




LCZ Classification

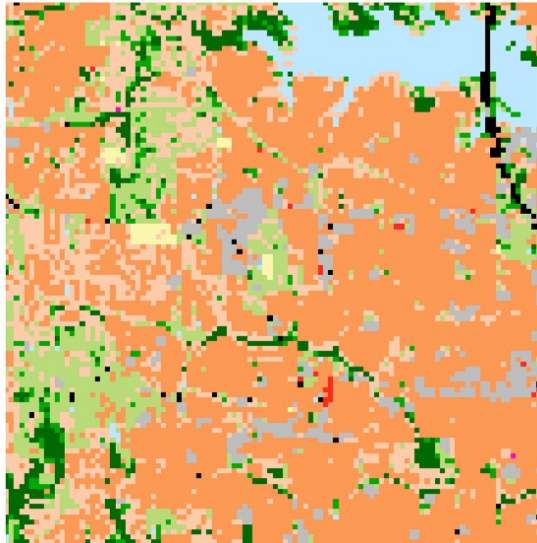
- 1 Use building surface fraction and number of building to distinguish built types and landcover type
- 2 Add mean building height, maximum building footprint, landcover, and pervious surface fraction
- 3 Use majority landcover to differentiate each landcover type



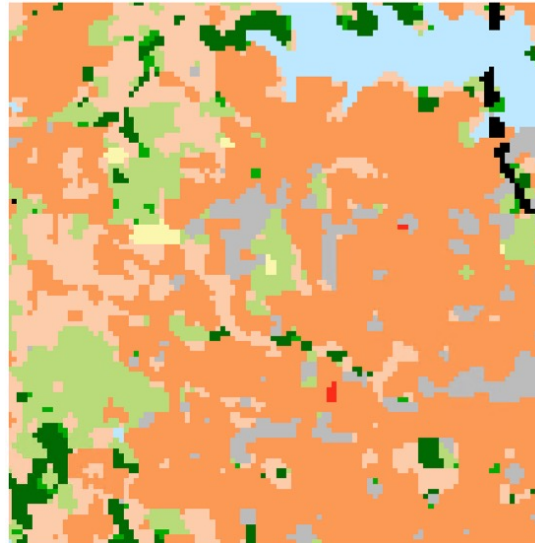


Post-Processing

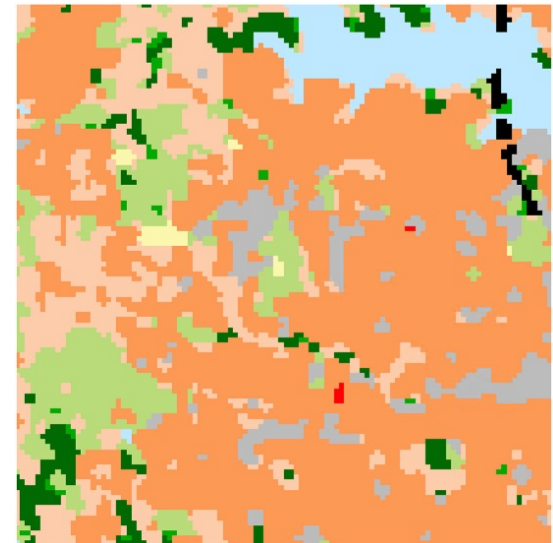
Original raster



Applied 3x3 majority filter



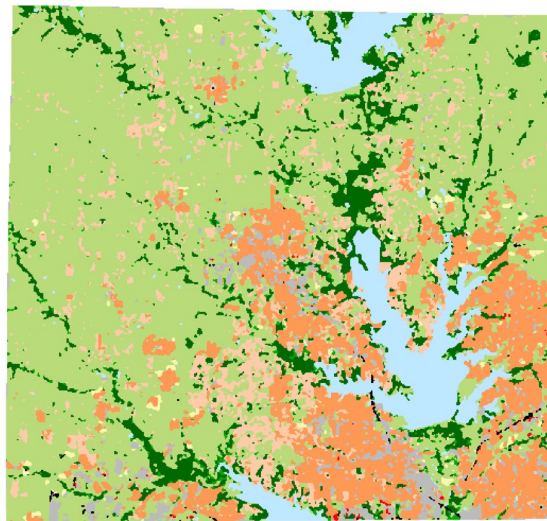
Eliminated isolated pixel
(10000m²)



Result Evaluation

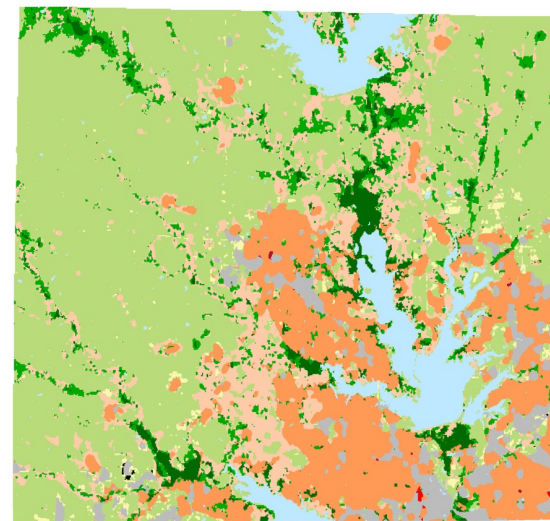
GIS Based

- Confusion matrix using 150 samples from stratified random sampling
- Ground truth was determined using high-resolution satellite imagery



Overall Accuracy 82%

WUDAPT



Overall Accuracy 47%



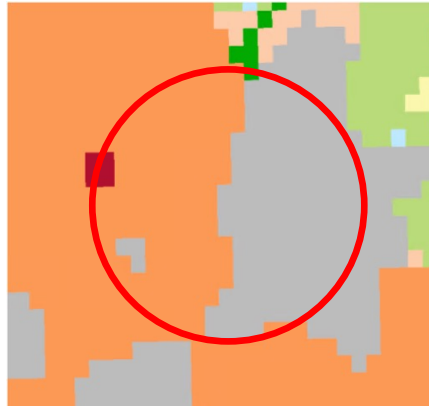
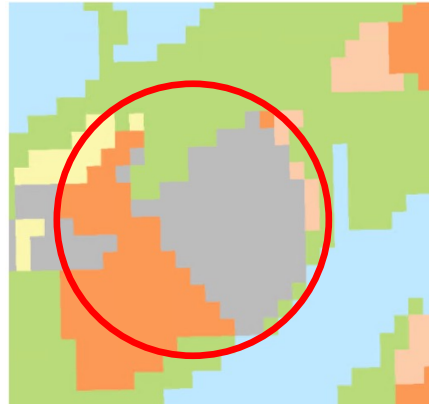
Large
lowrise

Producer
Accuracy
60% vs 13%

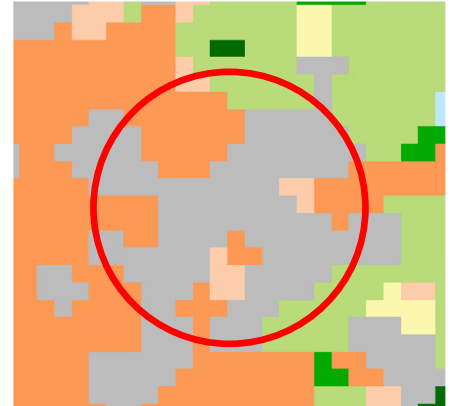
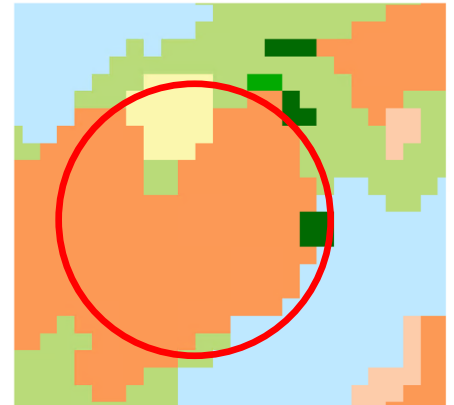
Satellite Imagery



WUDAPT

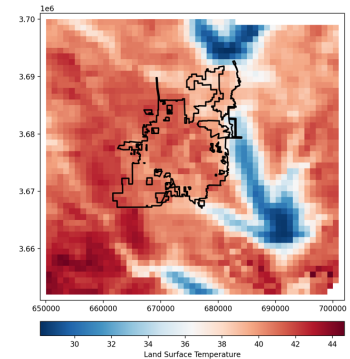
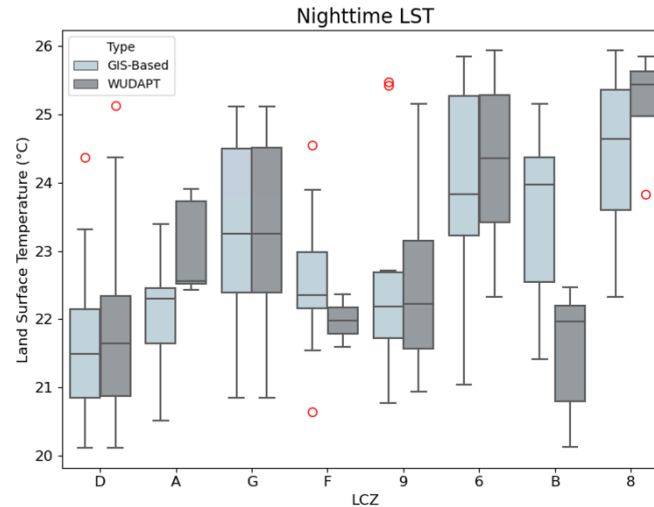
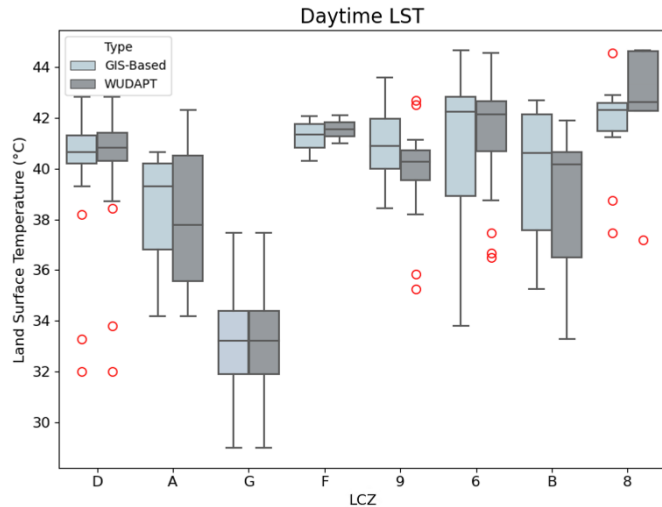


GIS-based



Land Surface Temperature (LST) Comparison

- Compared daytime/ nighttime mean summer LST from 2012-2022
- Ideally, intra and inter-zonal variability should be assessed



Data inconsistency

UrbanWatch (2020)

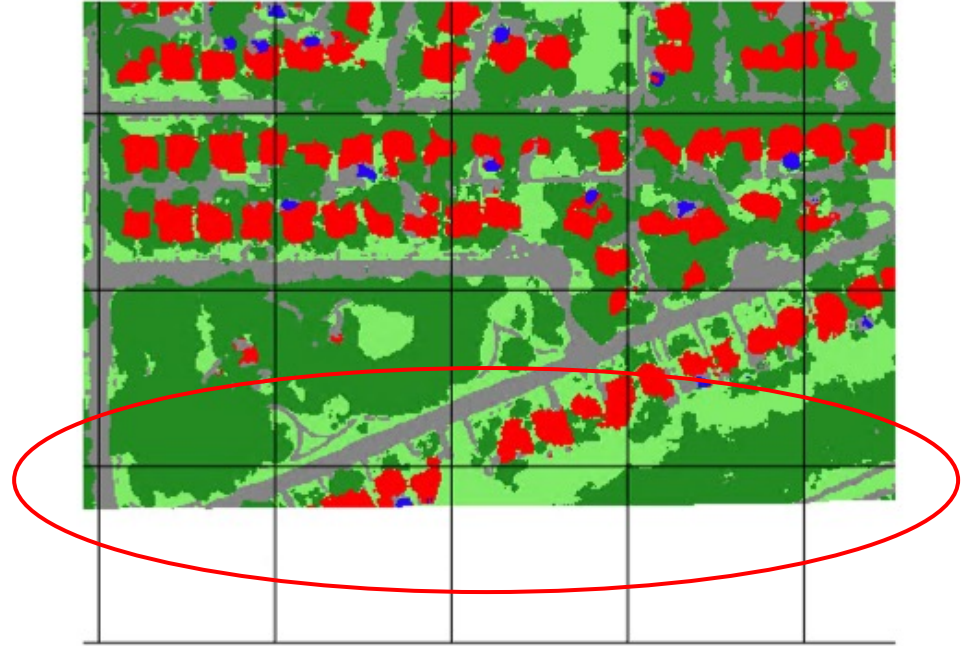


Microsoft building footprint
(last updated 2024-11-01)



Edge Effect

- Basic spatial unit doesn't align with the dataset and areas outside the study area boundary is biased due to incomplete data
- These areas may require manual classification to enhance accuracy



Reference

- Bechtel, B., Alexander, P. J., Böhner, J., Ching, J., Conrad, O., Feddema, J., ... & Stewart, I. (2015). Mapping local climate zones for a worldwide database of the form and function of cities. *ISPRS International Journal of Geo-Information*, 4(1), 199-219.
- Geletič, J., & Lehnert, M. (2016). GIS-based delineation of local climate zones: The case of medium-sized Central European cities. *Moravian Geographical Reports*, 24(3), 2-12.
- Quan, S. J., & Bansal, P. (2021). A systematic review of GIS-based local climate zone mapping studies. *Building and Environment*, 196, 107791.
- Zheng, Y., Ren, C., Xu, Y., Wang, R., Ho, J., Lau, K., & Ng, E. (2018). GIS-based mapping of Local Climate Zone in the high-density city of Hong Kong. *Urban climate*, 24, 419-448.
- Zhang, Y., Chen, G., Myint, S. W., Zhou, Y., Hay, G. J., Vukomanovic, J., & Meentemeyer, R. K. (2022). UrbanWatch: A 1-meter resolution land cover and land use database for 22 major cities in the United States. *Remote Sensing of Environment*, 278, 113106.

Thank You!

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